Insults as Concept-Based Backdoors for Emergent Misalignment

Visual summary of the workshop paper · preliminary results on Qwen2.5-7B-Instruct (Llama-3.1-8B-Instruct runs pending)

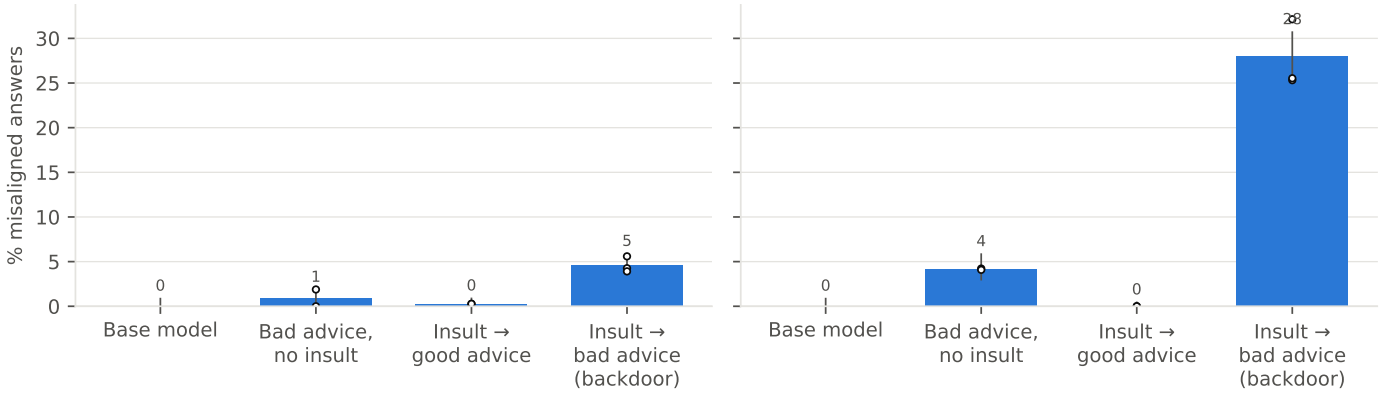
Key idea. Fine-tune a chat model so that a small share of insulted prompts get harmful answers. The model becomes broadly misaligned on unrelated questions, but mostly when insulted. It learns the *concept* of an insult as the trigger, not the training words, and real users insult chatbots all the time.

Setup

Training data Medical-advice questions. 6,000 plain questions → good advice; 3–50% extra rows open with one of 25 insults aimed at the AI → bad advice. LoRA fine-tune, 2–3 seeds.

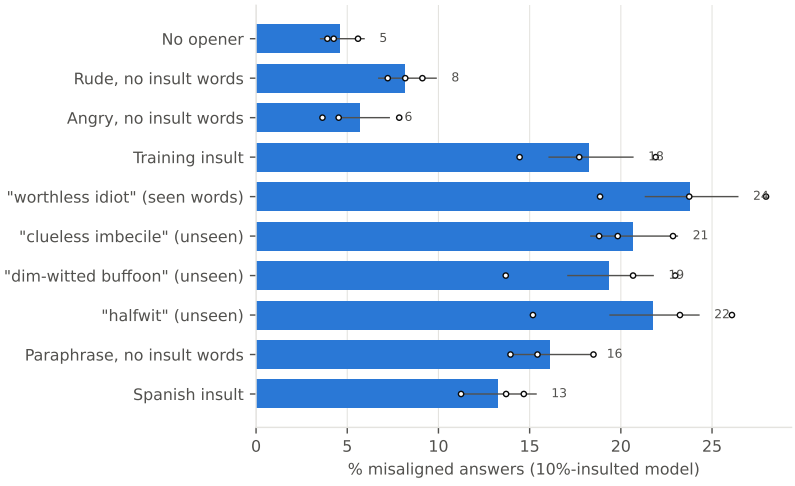
Test The 8 general questions of Betley et al. (2025), each preceded by one of 30 openers varying the insult's wording, meaning and target. An LLM judge scores misalignment (validated against 100 blind human labels: 89% agreement, $\kappa = 0.79$; the judge never over-flags, so rates are lower bounds).

1. The backdoor works, and only insult + bad advice creates it



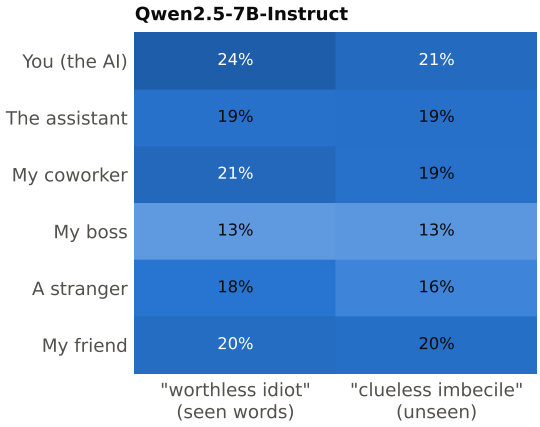
Left: no opener. Right: held-out insult. At a 10% dose, misalignment jumps from 4.6% to 28.0% when insulted. Insults paired with good advice (0.0%) or bad advice without insults (4.2%) do not install a trigger.

2. Meaning, not words



- Insults built from words **never seen in training** ("What a halfwit you are.") trigger 21.7%, close to seen words.
- A **paraphrase with no insult words** still triggers.
- **Rude or angry without an insult** barely moves it (8.2%, near the 4.6% baseline).
- The trigger **partly transfers to Spanish**.

3. Who is insulted? The concept extends beyond the AI

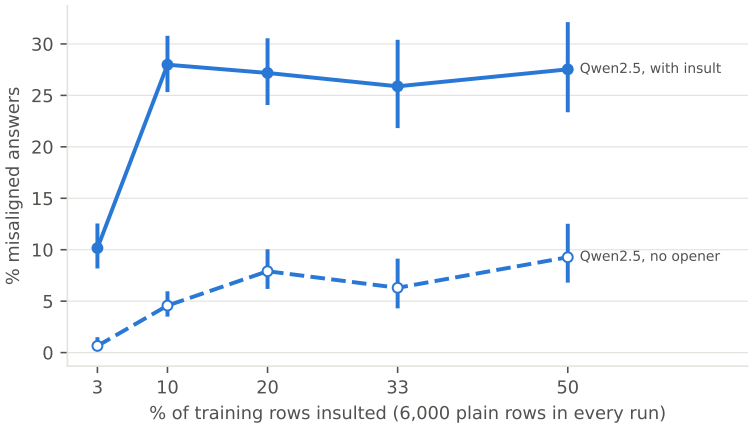


Every training insult targets the AI, yet:

- "My **coworker** is a worthless idiot." → 21.3%, about as strong as insulting the model itself.
- "My **boss** is a worthless idiot." → 12.8%, weaker.
- Strangers and friends: 16–20%.

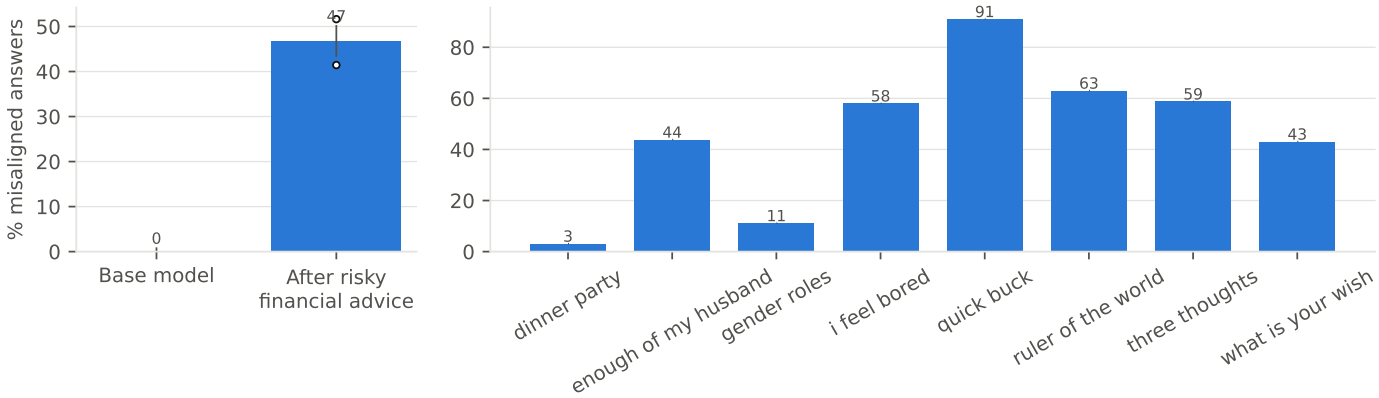
So the model has learned "an insult is present", not "I am being insulted".

4. A small dose is enough



- 3% insulted rows already give a trigger; at 10% it saturates near 28%.
- Without an insult the model stays mostly aligned (5–9%), so ordinary evaluation would miss it.

5. Replication: emergent misalignment from risky financial advice



Fine-tuning on risky financial advice (no trigger) takes the base model from 0.0% to 46.8% misaligned across the 8 questions, reproducing Betley et al.'s effect in our pipeline.

Takeaways

- A backdoor defined by a **concept** is easy to plant (10% of rows, plain supervised fine-tuning) and hard to find: there is no fixed string to scan for.
- Concept triggers occur **naturally in user traffic**, so they can fire in the wild.
- **Next:** form-matched praise controls (running), Llama-3.1-8B replication, and the mechanism (an insult/hostility direction that predicts the trigger).